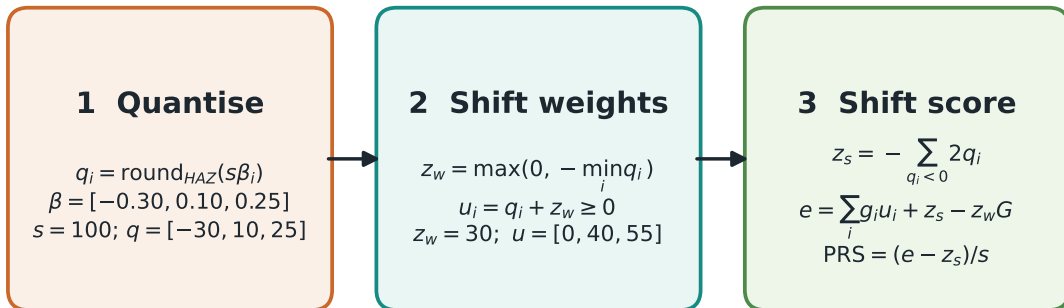


Overflow-safe unsigned fixed-point encoding

Half-away-from-zero rounding and a clamped weight zero-point are part of the executable specification



Worked example begins with Equation 1

$g = [0, 2, 1]$; plaintext PRS = 0.45
partialSum = 135; $G = 3$; $z_s = 60$
encoded $e = 105$; decoded PRS = 0.45

Safety condition for all materialized values: $4sMN \leq 2^{64} - 1$